Math 213 Final Exam Collaborative Review

Instructor Notes and Blackboard Setup

Purpose of the Review

The goal of today's class is not for students to watch the instructor solve every problem. The goal is for students to identify what they need to strengthen, practice explaining mathematics, ask useful questions, and help one another prepare for the final exam.

This format also allows the instructor to circulate, listen for misconceptions, make connections between topics, and answer targeted questions.

Because the final is cumulative, students should focus especially on recognizing *which idea or procedure applies* and explaining why.

What to Say at the Beginning of Class

Before we get started today, since this is our final review, I want to take a minute to step away from the exam itself and think about what you've actually gained from this class.

The reality is that some of you may use linear algebra a lot in your future classes or careers, and some of you may never calculate another determinant or find another eigenvector after this final. And that's okay. I don't expect the most important thing you take from this course to be that, ten years from now, you can remember the formula for an orthogonal projection.

But I do hope you take with you some of the skills and tools that you've developed while learning it.

Over the course of the semester, you've been given problems where you didn't immediately know what to do. You've had to look at the information you were given, decide which tools applied, break a complicated problem into smaller steps, recognize when something had gone wrong, go back and correct it, and explain why your answer made sense.

Those are useful skills whether or not the problem in front of you ever involves a matrix again.

"Whatever principle of intelligence we attain unto in this life, it will rise with us in the resurrection."
— D&C 130:18

This reminds me that education is about more than accumulating facts that we can reproduce on an exam. What we learn changes what we're capable of doing. Even when we eventually forget some of the details, we can carry forward the ability to reason carefully, to persist when something is difficult, to recognize patterns, to learn new things, and to use the tools available to us to solve problems.

So as you prepare for the final, of course I want you to review eigenvalues and determinants and projections and all of the mathematics we've learned. But I also hope you recognize how much more capable you are of approaching an unfamiliar problem now than you were at the beginning of the semester.

Today is an opportunity to see what tools you already have, identify the ones you still need to sharpen before the final, and help each other do that. As you move around the boards, don't just ask, "Do I remember how to do this problem?" Ask yourself, "How do I know what tool to use here, and could I explain that decision to someone else?"

If you can do that, you're doing much more than preparing for an exam.

Slide to project to students:

What Will You Take With You?

You may never calculate another determinant or find another eigenvector after this course.
But the goal is bigger than remembering formulas.

Think about the skills and tools you have learned.

You have practiced how to:

- Approach a problem when you do not immediately know what to do.
- Decide which tools and strategies apply.
- Break a complicated problem into smaller steps.
- Recognize when something has gone wrong and adjust your approach.
- Look for patterns and connections between ideas.
- Explain your reasoning clearly to someone else.

"Whatever principle of intelligence we attain unto in this life, it will rise with us in the resurrection."

– Doctrine and Covenants 130:18

What we learn changes what we're capable of doing.

Today's Goal:

Use the tools you have developed, identify the ones you still need to sharpen, and help one another prepare for the final.

Don't just ask: "Can I do this problem?"

Ask: "How do I know what tool to use, and can I explain why?"

Directions to Put on the Front Board

Collaborative Review Instructions

1. Choose the board containing the topic you most need to review.
2. Introduce yourself to the students at that board.
3. **Begin with the student prompts before doing the practice problem.**
4. Take turns explaining ideas and procedures. Do not let one person do all the work.
5. Ask questions whenever a step, definition, or strategy is unclear.
6. Check one another's work and identify common mistakes.
7. Move to another board once you feel more comfortable with the topic.
8. If you feel confident, help someone else before moving on.

Goal: Leave class knowing what you understand, what still needs practice, and what you will study before the final.

Board Topics to Write on Front Board

1. Systems, Invertibility, and Elementary Matrices
2. Subspaces, Span, Basis, and Linear Independence
3. Matrix Operations and Linear Transformations
4. Determinants
5. Eigenvalues, Eigenvectors, and Eigenspaces
6. Similarity and Diagonalization
7. Orthogonality, Projections, and Gram–Schmidt
8. Orthogonal Diagonalization and SVD

Note: Vectors and Geometry does not have its own station. Students who want additional practice with vectors, dot products, lines/planes, or geometric vector addition can use the Vectors and Geometry section of the Final Review independently.

Board 1: Systems, Invertibility, and Elementary Matrices

Skills:

- Interpret the RREF of an augmented matrix.
- Determine whether a system has no solution, one solution, or infinitely many solutions.
- Find rank and nullity.
- Use the Invertible Matrix Theorem.
- Construct elementary matrices and identify their row operations.
- Find the inverse of an elementary matrix by reversing the row operation.

Important Facts:

$$\text{rank}(A) = \text{number of pivots}, \qquad \text{nullity}(A) = \text{number of free variables},$$

$$\text{rank}(A) + \text{nullity}(A) = n \qquad (A \text{ is } m \times n).$$

Review the Invertible Matrix Theorem.

An **elementary matrix** is obtained by performing exactly one elementary row operation on I. If E represents a row operation, then EA performs that same row operation on A.

Row operation	**Inverse row operation**
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$
$R_i \to kR_i$	$R_i \to \frac{1}{k}R_i$
$R_i \to R_i + kR_j$	$R_i \to R_i - kR_j$

Student prompts:

- How do you know if a system is inconsistent?
- How do you identify free variables from RREF? What does nullity count?
- What are all the equivalent conditions for a matrix being invertible?
- How do you construct an elementary matrix from a row operation?
- If E represents a row operation, what does EA do?
- How can you find E^{-1} without row reducing E?
- What is the inverse operation for each of the three elementary row operations?

Practice:
Final Review — Section 2 # 1–2 and Section 3 # 2

Board 2: Subspaces, Span, Basis, and Linear Independence

Skills:

- Determine whether a set is a subspace.
- Determine whether vectors form a basis.
- Distinguish spanning from linear independence.
- Use quick tests for dependence and independence.

Important Facts:

A subset $S \subseteq \mathbb{R}^n$ is a subspace if

- $\mathbf{0} \in S$,
- S is closed under addition,
- S is closed under scalar multiplication.

$$\boxed{\text{Basis} = \text{linearly independent} + \text{spans the space.}}$$

Strategy for checking if n vectors are a basis for $\mathbb{R}^n$

For exactly n vectors in $\mathbb{R}^n$,

$$\text{LI} \iff \text{span } \mathbb{R}^n \iff \det(A) \neq 0 \iff A \text{ is invertible.}$$

Student prompts:

- What are the two requirements for a basis?
- If you have exactly n vectors in $\mathbb{R}^n$, what are several ways to check whether they form a basis?
- What clues let you rule out a basis without row reducing?
- Which subspace condition is usually easiest to check first?

Practice:
Final Review — Section 4 # 1–3

Board 3: Matrix Operations and Linear Transformations

Skills:

- Solve matrix equations while preserving multiplication order.
- Use transpose and inverse properties correctly.
- Find the standard matrix of a linear transformation.
- Compose linear transformations in the correct order.

Important Facts:

$$(AB)^T = B^T A^T, \qquad (AB)^{-1} = B^{-1} A^{-1}.$$

Matrix multiplication is generally not commutative: $AB \neq BA$.

The standard matrix of $T : \mathbb{R}^n \to \mathbb{R}^m$ is

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}.$$

If $T(\mathbf{x}) = A\mathbf{x}$ and $S(\mathbf{x}) = B\mathbf{x}$, then

$$(S \circ T)(\mathbf{x}) = S(T(\mathbf{x})) = BA\mathbf{x}.$$

Student prompts:

- When solving a matrix equation, why does it matter which side you multiply on?
- Why does the order reverse in $(AB)^T$ and $(AB)^{-1}$?
- How do you find the standard matrix from a description of a transformation T?
- What are the standard matrices for common rotations, reflections, projections, or scaling?

Practice:
Final Review — Section 3 # 1 and Section 5 #1–2

Board 4: Determinants

Skills:

- Compute determinants efficiently.
- Choose a useful row or column for cofactor expansion.
- Use determinant properties.
- Track the effect of row operations on the determinant.
- Connect determinants with invertibility.

Important Facts:

$$\det(AB) = \det(A)\det(B), \qquad \det(A^T) = \det(A), \qquad \det(A^{-1}) = \frac{1}{\det(A)}.$$

For an $n \times n$ matrix,

$$\boxed{\det(kA) = k^n \det(A).}$$

Effect of row operations:

- $R_i \leftrightarrow R_j$: determinant changes sign.
- $R_i \to kR_i$: determinant is multiplied by k.
- $R_i \to R_i + kR_j$: determinant is unchanged.

For a triangular matrix,

$$\det(A) = \text{product of the diagonal entries.}$$

Student prompts:

- Before calculating a determinant, what structure should you look for?
- Which row or column should you choose for cofactor expansion?
- What effect does each elementary row operation have on the determinant?
- Why is $\det(kA) = k^n \det(A)$ rather than $k\det(A)$?
- What does $\det(A) = 0$ tell you about A?

Practice:
Final Review — Section 6 #1–2

Board 5: Eigenvalues, Eigenvectors, and Eigenspaces

Skills:

- Find eigenvalues from the characteristic equation.
- Find eigenspaces and bases for eigenspaces.
- Use triangular matrices efficiently.
- Determine algebraic and geometric multiplicities.

Important Facts:

$$A\mathbf{v} = \lambda\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}.$$

Eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

For an eigenvalue λ, its corresponding eigenvectors are the **nonzero** vectors in

$$E_\lambda = \mathrm{Null}(A - \lambda I).$$

$$\mathrm{AM}(\lambda) = \text{multiplicity of } \lambda \text{ as a root of } p_A(\lambda),$$

$$\mathrm{GM}(\lambda) = \dim(E_\lambda), \qquad \boxed{1 \leq \mathrm{GM}(\lambda) \leq \mathrm{AM}(\lambda).}$$

For a triangular matrix, the eigenvalues are the diagonal entries.

Student prompts:

- How do you find the eigenvalues and eigenvectors of a matrix?
- How do you find a basis for an eigenspace?
- Where does algebraic multiplicity come from? Where does geometric multiplicity come from?
- What inequality always relates AM and GM?
- What are the eigenvalues of a triangular matrix?

Practice:
Final Review — Section 7 # 1–3

Board 6: Similarity and Diagonalization

Skills:

- Determine whether matrices could be similar.
- Determine whether a matrix is diagonalizable.
- Use algebraic and geometric multiplicities to test diagonalizability.
- Understand how P and D are constructed in $A = PDP^{-1}$.

Important Facts:

Matrices A and B are similar if $P^{-1}AP = B$ for some invertible matrix P.

Similar matrices have the same characteristic polynomial, eigenvalues, trace, and determinant.

Equivalent Conditions for Diagonalizability

For an $n \times n$ matrix A, the following are equivalent:

1. A is diagonalizable.
2. A has n linearly independent eigenvectors.
3. There is a basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A.
4. For every eigenvalue, geometric multiplicity = algebraic multiplicity.
5. The sum of the dimensions of all eigenspaces is n.

If A has n distinct eigenvalues, then A is diagonalizable. The converse is not necessarily true.

Student prompts:

- What does $A \sim B$ mean? What information is preserved under similarity?
- Which quantities can quickly rule out similarity?
- Does a repeated eigenvalue automatically mean a matrix is not diagonalizable?
- What goes in the columns of P? What goes on the diagonal of D?

Practice:
Final Review — Section 8 # 1–2

Board 7: Orthogonality, Projections, and Gram–Schmidt

Skills:

- Determine whether vectors are orthogonal or orthonormal.
- Find a basis for an orthogonal complement.
- Project a vector onto a subspace with an orthogonal basis.
- Apply the Gram–Schmidt process.

Important Facts:

$$\mathbf{u} \text{ orthogonal to } \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

If $W = \mathrm{Col}(A)$, then

$$\boxed{W^{\perp} = \mathrm{Null}(A^T).}$$

Projection onto a nonzero vector:

$$\mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}.$$

If $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an orthogonal basis for W, then

$$\mathrm{proj}_W(\mathbf{v}) = \sum_{i=1}^{k} \frac{\mathbf{v} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}\mathbf{u}_i.$$

Gram–Schmidt:

Takes a basis $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ and orthogonalizes it:

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1, \\
\mathbf{v}_2 &= \mathbf{x}_2 - \mathrm{proj}_{\mathbf{v}_1}(\mathbf{x}_2), \\
\mathbf{v}_3 &= \mathbf{x}_3 - \mathrm{proj}_{\mathbf{v}_1}(\mathbf{x}_3) - \mathrm{proj}_{\mathbf{v}_2}(\mathbf{x}_3).
\end{aligned}$$

Student prompts:

- What is the difference between orthogonal and orthonormal?
- How can you find a basis for $W^{\perp}$?
- How do you find the projection of a vector onto a given subspace $\mathrm{proj}_W(\mathbf{v})$?
- How does the projection formula simplify for an orthonormal basis?
- What does gram-Schmidt do to a given basis?

Practice:
Final Review — Section 9 # 1–3

Board 8: Orthogonal Diagonalization and SVD

Skills:

- Recognize orthogonal matrices.
- Orthogonally diagonalize a symmetric matrix.
- Apply the Spectral Theorem.
- Find singular values.
- Construct a singular value decomposition.

Important Facts:

For an orthogonal matrix Q,

$$Q^T Q = I, \qquad Q^{-1} = Q^T, \qquad \det(Q) = \pm 1.$$

The columns of Q form an orthonormal basis.

Spectral Theorem

A real matrix A is orthogonally diagonalizable if and only if A is symmetric:

$$\boxed{A = A^T \iff A = QDQ^T \text{ for some orthogonal } Q.}$$

The columns of Q are orthonormal eigenvectors of A.

Look at the roadmap for finding the Singular Value Decomposition of a matrix.

Student prompts:

- How can you recognize an orthogonal matrix from its columns?
- Which matrices are orthogonally diagonalizable?
- How is orthogonal diagonalization different from ordinary diagonalization?
- If an eigenvalue is repeated, what must be true about the basis chosen for that eigenspace before putting it into Q?
- How do you find the singular values of a matrix?
- What are the steps to find the SVD of a matrix?

Practice:
Final Review — Section 10 #1–2, Section 11 # 1–2